

INTELLIGENT AUTOMATION

AI Configurator

Automated electrical panel pre-configuration — from customer request to bill of materials in seconds.

Today's Pre-Configuration Process



Time-consuming

Engineers manually browse 2 600+ components to build each configuration.



Knowledge-intensive

Matching the right enclosure and components requires deep product expertise.



Repetitive work

Similar projects are re-created from scratch each time without leveraging past experience.

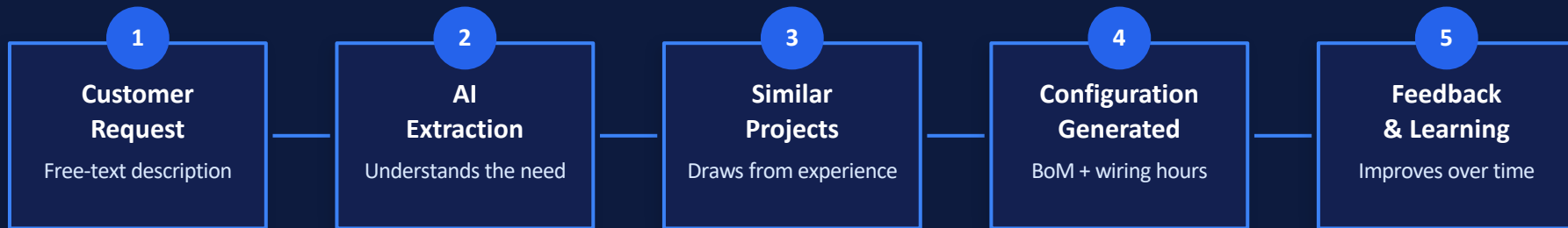


Documentation overhead

Generating BoM, wiring estimates, and pricing takes additional manual effort.

From Request to Configuration in One Step

The CETIE AI Configurator combines a large language model with your company's historical project database to instantly generate accurate, validated pre-configurations — and it gets smarter with every correction.



↻ *Feedback loop — every correction improves future configurations*

Enter Your Request — AI Does the Rest

1 Enter the Customer Request

"Armoire commande 2 pompes relevage eaux usées 7.5 kW, protection IP65, secteur industrie agro-alimentaire"

Any language · Any format · Plain text description

Generate Configuration

2 AI Analyses & Clarifies

Product type	Motor control enclosure
--------------	-------------------------

Power	7.5 kW × 2 motors
-------	-------------------

Protection	IP65
------------	------

Sector	Agri-food
--------	-----------

Application	Wastewater pumping
-------------	--------------------

 *If critical information is missing, the AI asks targeted clarification questions before proceeding.*

Configuration Generated Automatically

Enclosure Selected

Atlantic 800×600×300 IP65
Polyester — 151 options evaluated

✓ *Best fit for 2 × 7.5 kW*

Time Estimates

⚡ Wiring hours **14 h**

⚙️ Automation hrs **8 h**

Bill of Materials

- Circuit breaker 3P 25A
- Contactor LC1D18
- Thermal relay LR2D
- IP65 cable glands ×6

Reference	Description	Justification	Qty	Unit (€)
GV3ME20	Motor circuit-breaker 18-25A	Motor protection 7.5kW	2	89.50
LC1D18	3-pole contactor 18A	Motor start/stop control	2	42.00
LR2D13	Thermal overload relay	Overcurrent protection	2	28.00

Compare with Similar Past Projects



Scores show how closely a past project matches the current request — green means highly similar.



Compare your estimated wiring hours against similar completed jobs to validate the estimate.



See the sector, product type, and brief solution summary for context.



Powered by semantic search across all historical CETIE quotes — not just keyword matching.



Similar Past Projects

96%

Pump control 2×7.5kW

Agri-food · 12h

88%

Motor panel IP65

Industry · 16h

74%

Control cabinet 15kW

Water · 18h



Feedback Loop — The System Learns



Configuration Looks Good

1. Click  Feedback in the right panel
2. Select "Looks Good"
3. System records this as a positive example
4. Future similar requests benefit from this validation



Make a Correction

1. Select "Needs Correction"
2. Describe what was wrong or what you changed
3. Correction is stored and injected into future prompts
4. Same mistake won't happen again for similar requests



Promote to Permanent Rule

1. From the  Rules tab, view all learned rules
2. Activate or deactivate rules at any time
3. Permanent rules apply to EVERY future configuration
4. Example: 'Always add a main switch for 3-phase 400V'

Under the Hood — Key Components



Language Model (AI)

Claude (Anthropic) reads the customer request, extracts structured data, chooses components from the CETIE catalogue, estimates wiring time, and generates justifications — all in one pass.



CETIE Component Database

2 661 components (BDD_Blocs) and 151 enclosures (BDD_Armoires) exported from your Excel files. The AI selects from this real catalogue — no hallucinated products.



Semantic Project Search

Historical quotes are indexed using vector embeddings (OpenAI). When a new request arrives, the system instantly finds the 10 most similar past projects to provide context and validation.



Feedback & Rules Engine

Every correction or rule added by your engineers is stored locally and injected into the next AI prompt. No retraining required — the system adapts in real time.

Local, Lightweight, Ready to Share



Runs locally

Flask web app on any PC. No cloud dependency for the UI or database — just API calls to AI services when needed.



Portable

The entire application (code + component database + historical project index) fits in a single ~10 MB zip file.



API keys required

Each team member needs their own OpenAI and Anthropic API keys. These stay in a local .env file — never committed to source control.



Persistent learning

Feedback and rules are stored in simple JSON files locally. Share the zip with your coworker and the learned rules travel with it.



Data stays internal

Customer request text is sent to AI APIs for processing — no other data leaves your machine. Historical quotes are stored locally.



Setup in minutes

```
pip install requirements · add .env · python app.py — that's it.
```

What This Delivers

< 30s

configuration time
(vs. 30–60 min manually)

2 600+

real components
always up-to-date

10×

similar past projects
surface automatically

- Engineers focus on engineering — not catalogue browsing
- All component references validated against the real CETIE catalogue
- Configurations improve automatically from team corrections
- Wiring and automation time estimates built into every output
- New team members produce expert-level pre-configs from day one
- Full auditability: justifications for every component selection

Next Steps

- 01 Run the POC on representative CETIE requests and collect feedback from the engineering team.
- 02 Enrich the historical project database with additional annotated quotes to improve similarity search accuracy.
- 03 Validate component selection against recent orders and calibrate the wiring hour estimates.
- 04 Define permanent rules for CETIE standards (standards, preferred brands, mandatory safety devices).